IMPORTANT **Definition of Thompson sampling**

The idea of Thompson sampling is to first, maintain a probabilistic belief about the quantity that we care about, and then sample from that belief and treat that sample as a replacement of the true unknown quantity in question. This sampled replacement is then used to pick out the best decision we should make.

Let's come back to our weather forecast example to see how this works. Again, we are interested in the problem of deciding whether we should bring an umbrella to work, where we are given an estimate of the probability that the weather willl stay sunny throughout the day. Remember that UCB will rely on an upper bound of this estimate to inform its decisions, but what will the TS policy do?

TS will first draw a sample from the probability distribution that we use to model the unknown quantity—whether it will stay sunny in this case—and make decisions based on this sample. Let's say instead of a range like what we saw in the previous section, the weather app on our phone now announces that there's a 66 percent (roughly two out of three) chance that the weather will say sunny. This means, by following the TS policy, that we will first flip a coin with a two-thirds bias for heads.

- If the coin lands on heads (with 66 percent chance), then we treat it as if the weather will stay clear throughout the day and conclude that we won't need an umbrella.
- If the coin lands on tails (with 34 percent chance), then we treat it as if it will rain and conclude we should bring an umbrella to work.

This process of TS is visualized in Figure 5.11 as a decision tree, where at the beginning, we flip a biased coin to obtain a sample from the probability distribution of sunny weather. Based on whether the coin lands on heads (representing a sample of sunny weather) or tails (a sample of rain), we will decide whether we should bring an umbrella to work or not.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>